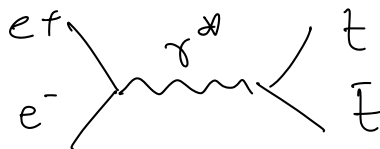


Top Quark

$$e^+e^- \rightarrow t\bar{t}$$

@ Fermilab 1994



$$\sqrt{s} \geq 2m_t$$

LEP @ CERN e^+e^- $\sqrt{s}, 90 \rightarrow 200$ GeV 1989-2001
no direct signal for t

$$\Rightarrow m_t \geq \frac{\sqrt{s}}{2} = 100 \text{ GeV}$$

$$p + p/\bar{p} \rightarrow t\bar{t} + X$$

(SPS @ CERN, Tevatron @ Fermilab)

From LEP and other indirect probes.



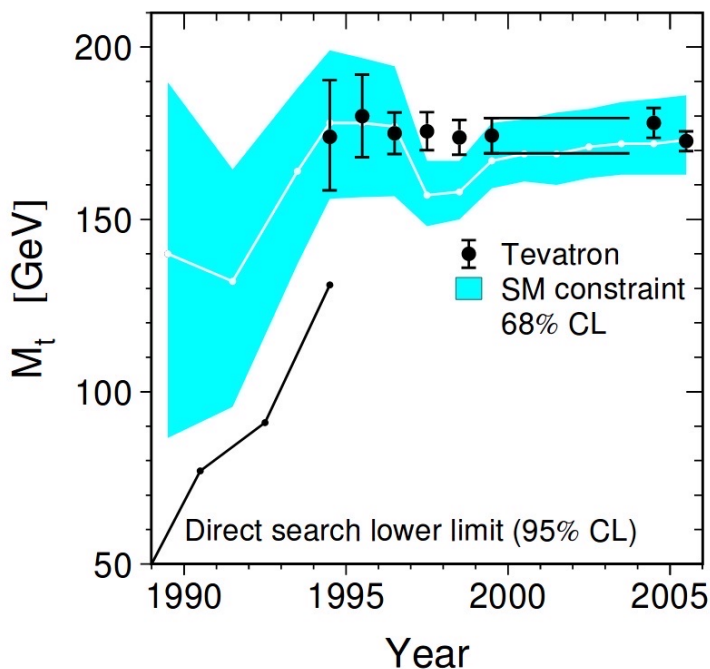
$$m_t > 100 \text{ GeV}$$

Tevatron $p\bar{p}$ $\sqrt{s} = 1.96$ TeV
Symmetric colliding beams

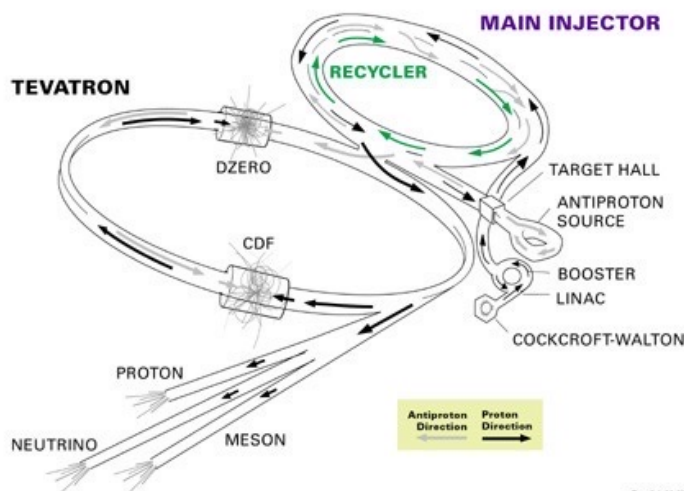
$$p \rightarrow \leftarrow \bar{p}$$

$$\Rightarrow E_{beam} = \frac{\sqrt{s}}{2}$$

$$p + p \rightarrow p + \bar{p} + p + p \Rightarrow \text{produce beam of } \bar{p}$$

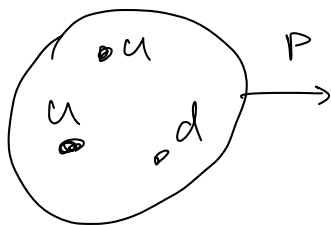


FERMILAB'S ACCELERATOR CHAIN



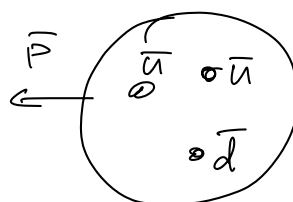


$$\sqrt{s} = 1.96 \text{ TeV} \Rightarrow \text{proton/antiproton} = E_{\text{beam}} \approx 1 \text{ TeV}$$



$$\underline{P}_1 = x_1 P$$

x_1 : fract. of mom.
carried by parton
in proton



$$\underline{P}_2 = -x_2 P$$

x_2 : fract. of mom.
carried by parton
in $\bar{P}$

$$\underline{P}_1 = (x_1 P, x_1 P, 0, 0)$$

negligible p mass

$$\underline{P}_2 = (x_2 P, -x_2 P, 0, 0)$$

$$S_{pp} = (\underline{P}_1 + \underline{P}_2)^2 = [(x_1 + x_2) P]^2 - [(x_1 - x_2) P]^2$$

even if $x_1 \neq x_2 \Rightarrow S_{pp} = (x_1 + x_2)^2 P^2$

$$\sqrt{S_{pp}} = (x_1 + x_2) P \approx 2xP$$

$$x_1 = x_2 = \frac{1}{3} \quad \sqrt{S_{pp}} = \frac{2}{3} P \approx 670 \text{ GeV}$$

1) parton-parton interaction $\sqrt{S_{pp}} < \sqrt{s} = 1.96 \text{ TeV}$

$$2) \quad x_1 \neq x_2 \quad \xrightarrow{x_1 P} \quad \xleftarrow{x_2 P} \quad \xrightarrow{\quad} z$$

$$\vec{P}_{in} = (x_1 - x_2) P \quad (x_1 - x_2) > 0.$$

$\Rightarrow$ Lorentz boost in $\vec{z}$ direction.

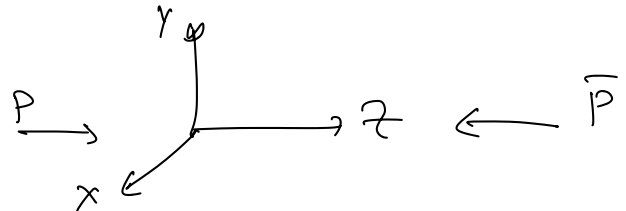
$$x_1 < x_2 \quad \xrightarrow{x_1 P} \quad \xleftarrow{x_2 P} \quad \xrightarrow{\quad} z$$

$$x_1 - x_2 < 0 \Rightarrow \text{boost in } -\vec{z}$$

$P_{tot\,x} \neq 0$ Different than e^+e^- symm. colliders.

Transverse plane :

$$\vec{P}_{T,in} = \vec{0}$$



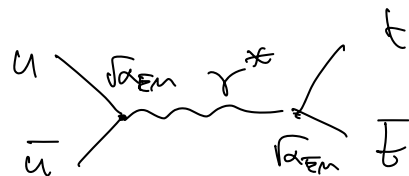
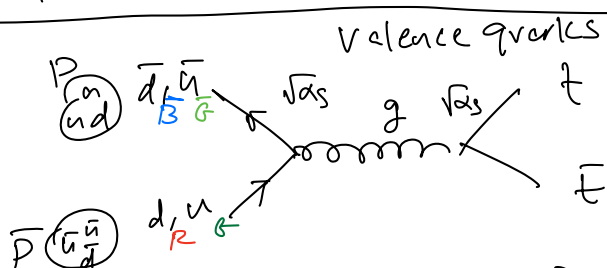
$$\vec{P}_{T,in} = (P_{x1} + P_{x2}) \hat{x} + (P_{y1} + P_{y2}) \hat{y} = \vec{0}$$

because beams along z direction.

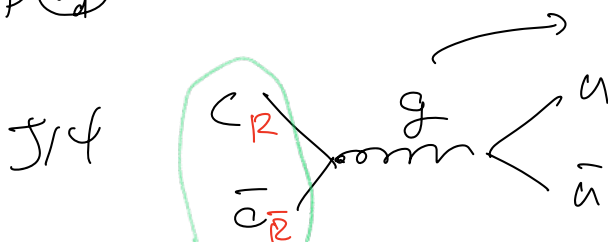
$$\Rightarrow \text{Final state } \vec{P}_{T,f.in} = \vec{0}$$

$\Rightarrow$ transverse mom. has to be conserved.

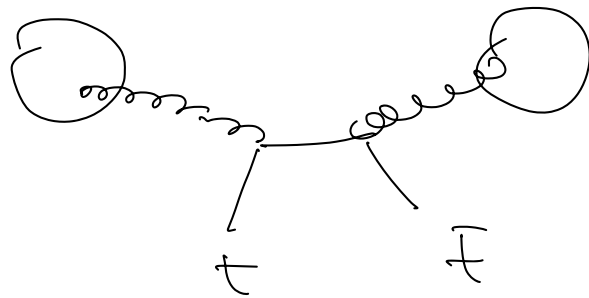
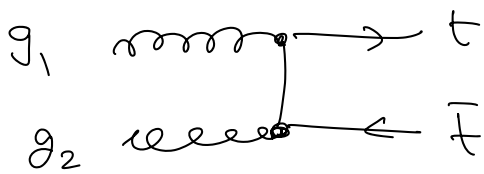
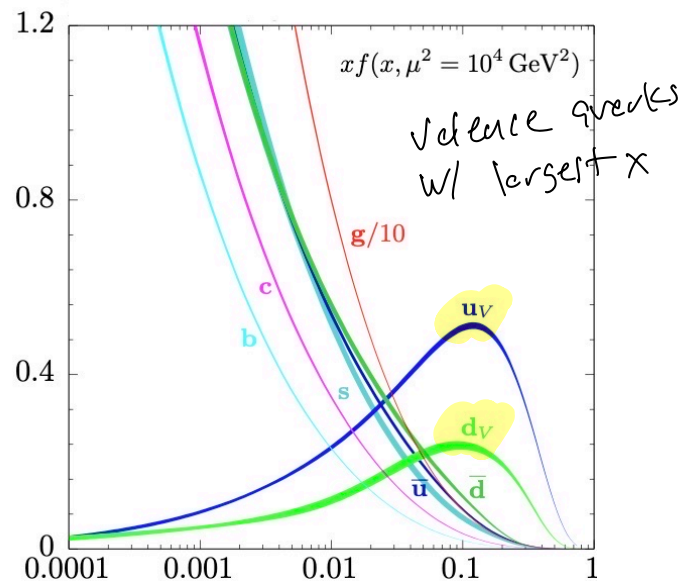
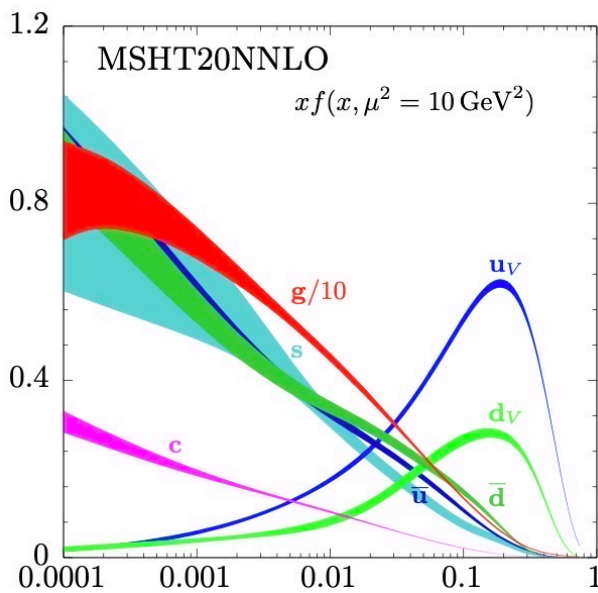
top production in $p\bar{p}$



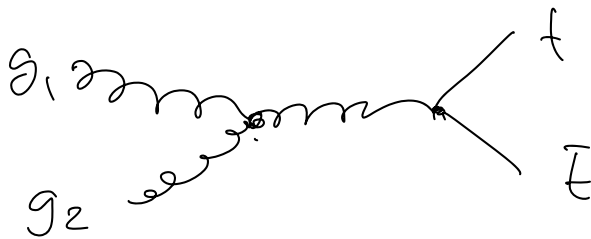
gluons are color octets



color
singlet



higher $\sqrt{s}$ provides access to small x gluons



gluon-gluon fusion

$$m_{\text{top}} \approx 174 \text{ GeV} \quad \Gamma \approx 2 \text{ GeV}$$

Toponium ($t\bar{t}$) exist?

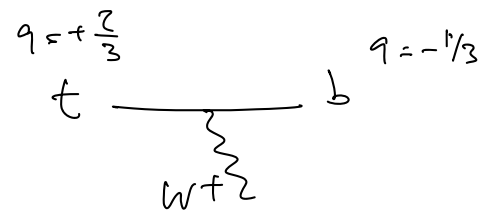
weak decay: $\Gamma(t \rightarrow ab) \sim G_F^2 m_t^3$

$$G_F \approx 1 \times 10^{-5} \text{ GeV}^{-2}$$

$\Rightarrow \tau = \frac{1}{\Gamma} \approx 10^{-25} \text{ sec.} \ll \text{time needed for QCD to form bound state } t\bar{t}$

Hint at CMS $t\bar{t}$ might exist (2024)

$\Rightarrow$ open question if toponium can exist

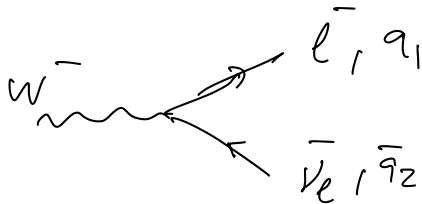
$$q = +\frac{2}{3} \quad q = -\frac{1}{3}$$




$$m_W = 80 \text{ GeV} < m_{top}$$

if no bound state formed.

$$p + p \rightarrow t + \bar{t} \rightarrow \begin{cases} \bar{b} + W^- \\ b + W^+ \end{cases}$$

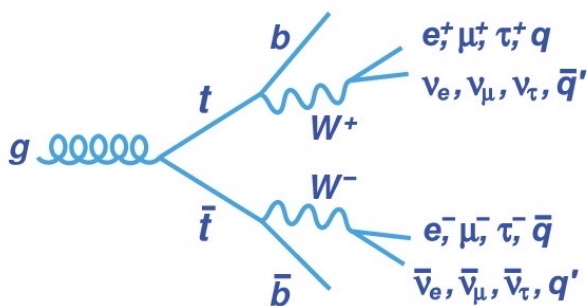


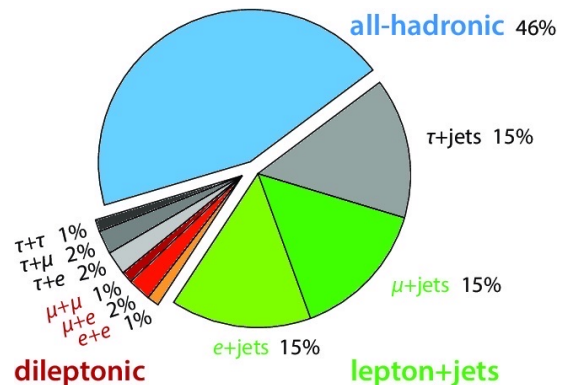
W⁺ DECAY MODES	Fraction (Γ_i/Γ)	Confidence level	P (MeV/c)
$\ell^+ \nu$	[b] $(10.86 \pm 0.09) \%$	—	—
$e^+ \nu$	$(10.71 \pm 0.16) \%$	40189	—
$\mu^+ \nu$	$(10.63 \pm 0.15) \%$	40189	—
$\tau^+ \nu$	$(11.38 \pm 0.21) \%$	40170	—
hadrons	$(67.41 \pm 0.27) \%$	—	—

$$p + p \rightarrow t + \bar{t} \rightarrow \begin{cases} b + \bar{b} + \ell^- + \bar{\nu}_\ell + \ell^+ + \nu_\ell \\ \vdots \end{cases}$$

$$b + \bar{b} + q_1 + \bar{q}_2 + q_3 + \bar{q}_4$$

6 hadronic jets.





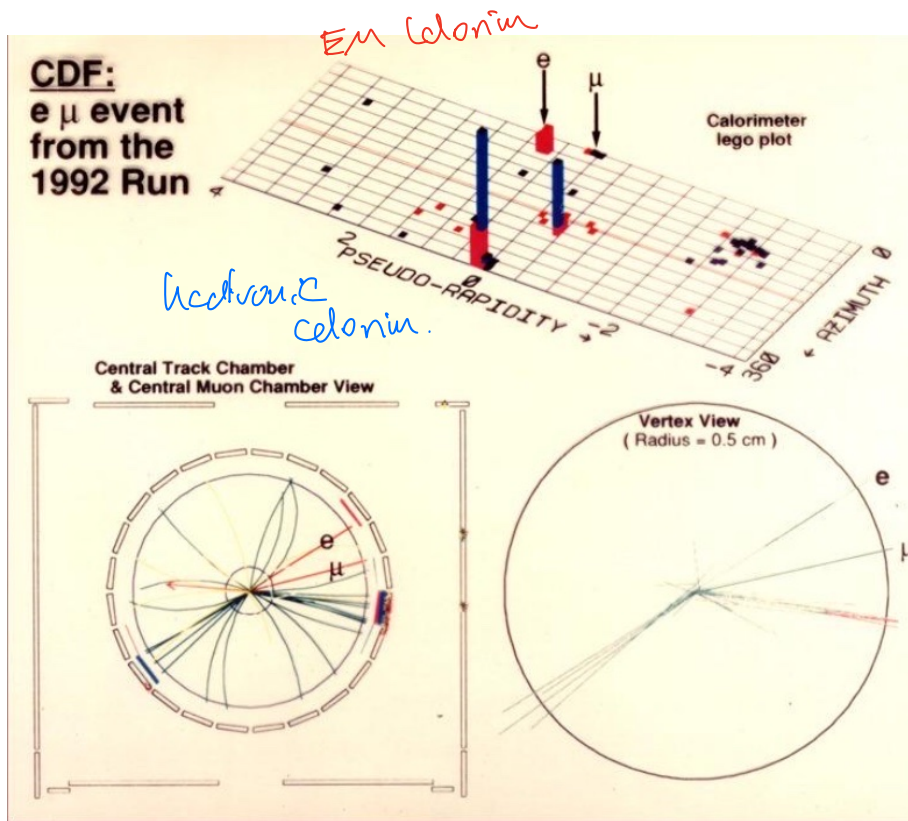
$$\mathcal{H}(t\bar{t} \rightarrow \bar{b}\bar{b}\ell\bar{\ell} + \bar{\nu}_\ell + \nu_\ell) = \sigma_{t\bar{t}} \times L_{inst} \times \Delta t \times$$

$$1 \simeq \text{BF}(t \rightarrow W) \times \text{BF}(\bar{t} \rightarrow W) \times$$

$$10\% \times 90\% \simeq 1\% \leftarrow \times \text{BF}(W \rightarrow \ell\nu) \times \text{BF}(W \rightarrow \ell\nu)$$

Golden mode: - small number of events.
 - exp. very clean.

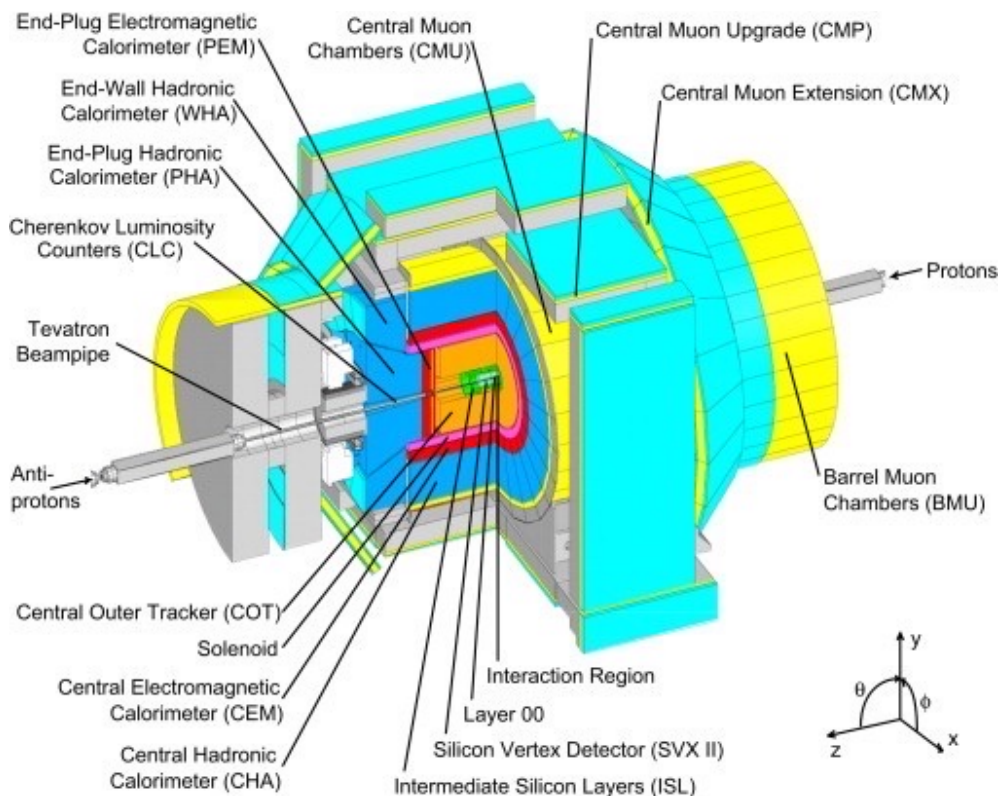
Exp. signature: 2 bjets + 2 leptons + missing transverse momentum.

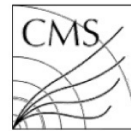


2 Detectors

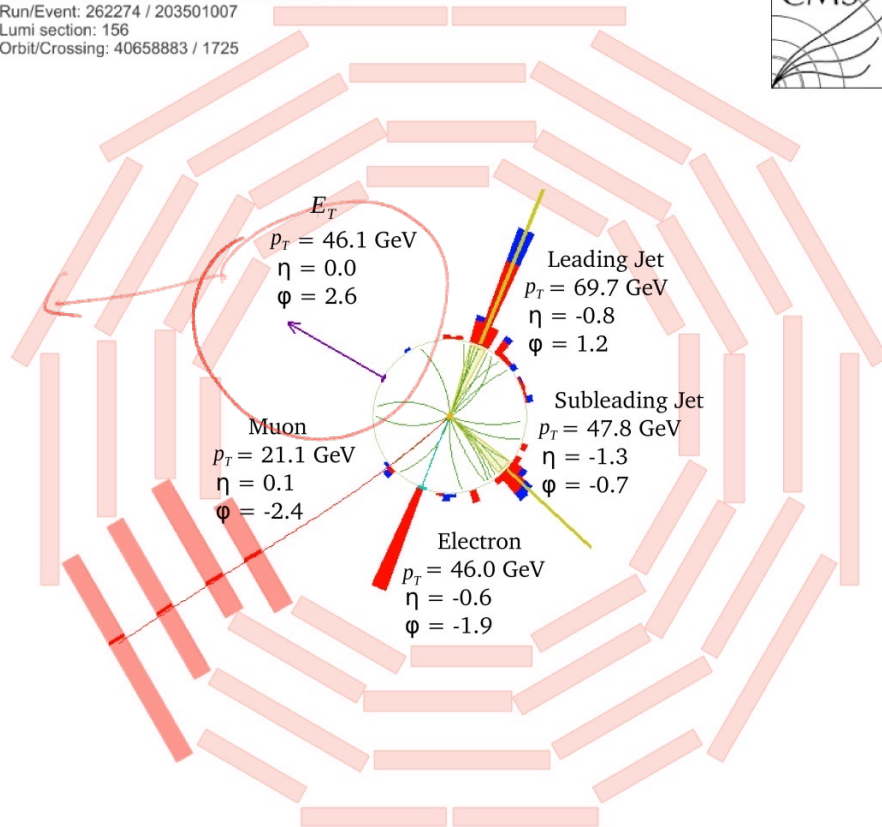
CDF

DØ





$\nu + \bar{\nu}$



Missing transverse momentum:

$$\vec{P}_{Tj_1} + \vec{P}_{Tj_2} + \vec{P}_{T\mu} + \vec{P}_{Te} \neq \vec{0}$$

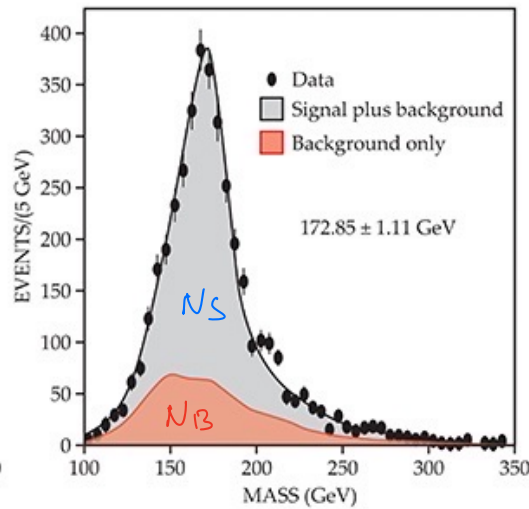
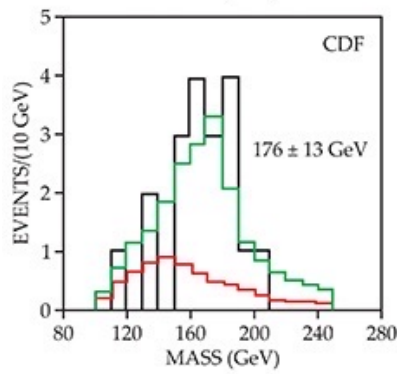
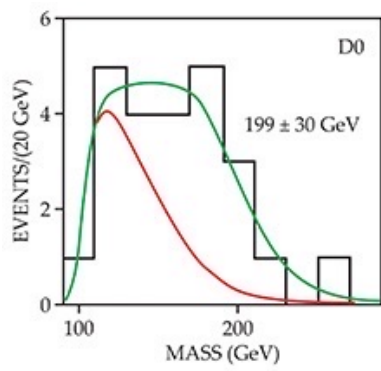
From conservation $\sum_i \vec{P}_{Ti} = \vec{0}$

$$\Rightarrow \vec{P}_{T\nu\bar{\nu}} = - \sum_{\text{meas. Mom.}} \vec{P}_{Ti}$$

$$MET = |\vec{P}_{T\nu\bar{\nu}}|$$

$t\bar{t} \rightarrow b\bar{b} + \ell^+ \ell^- + \text{missing energy. (MET)}$

$$M_{inv} = \sqrt{(\vec{P}_b + \vec{P}_\ell + \vec{P}_\nu)^2}$$



$$\hat{N}_S = N_{\text{total}} - N_B$$

$$S.S = \frac{\hat{N}_S}{\sqrt{N_B}}$$

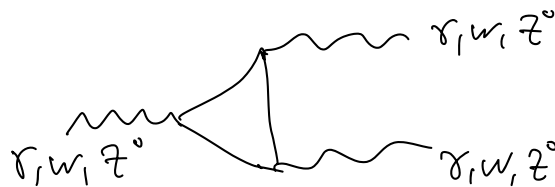
$$\begin{pmatrix} u \\ d \end{pmatrix} \quad \begin{pmatrix} c \\ s \end{pmatrix} \quad \begin{pmatrix} t \\ b \end{pmatrix}$$

$$\begin{pmatrix} e^- \\ \nu_e \end{pmatrix} \quad \begin{pmatrix} \mu^- \\ \nu_\mu \end{pmatrix} \quad \begin{pmatrix} \tau^- \\ \nu_\tau \end{pmatrix}$$

1st 2nd 3rd

ν_τ : DONUT experiment in 1999

Standard Model each family must be complete
If SM correct

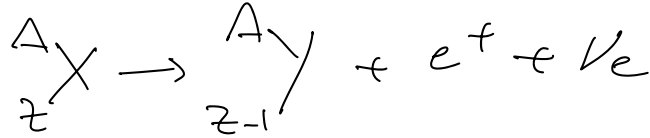
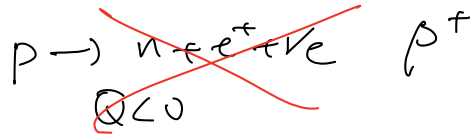
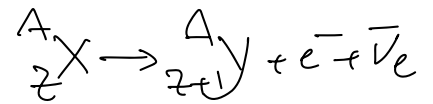
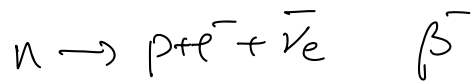


anomalies
in SM.

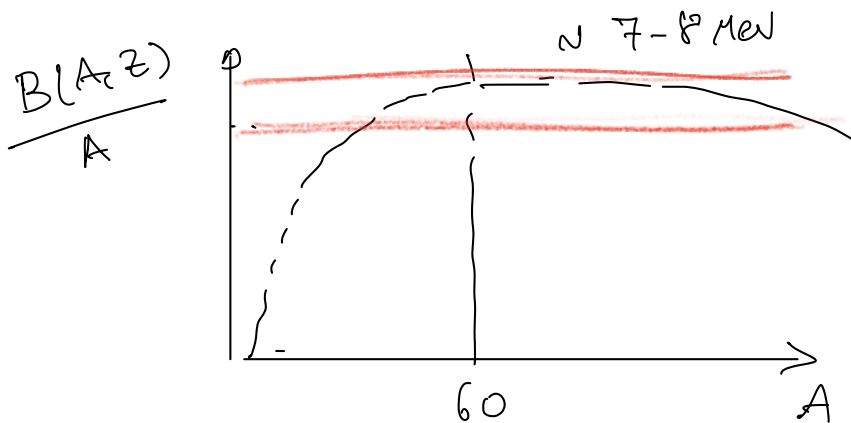
If families are complete $\Rightarrow \sum_i \text{anomalies} = 0$
cancellation of
anomalies

Weak Interaction

β decays



$$M(A, Z) = (A - Z) m_n + Z \cdot m_p - B(A, Z)$$



Bethe-Weizsäcker Formula.